

Kosuke Ukita

PhD student (D1), full-time

Kyushu Institute of Technology

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EDUCATION

Apr. 2026 - Present	PhD Kyushu Institute of Technology Department of Computer Science and Systems Engineering, Graduate School of Computer Science and Systems Engineering
Apr. 2024 - Mar. 2026	MSc Kyushu Institute of Technology Department of Creative Informatics, Graduate School of Computer Science and Systems Engineering <u>Thesis: "A Study on Sensor Foundation Models via Joint Training using Flow Matching"</u> GPA: 3.6 / 4.0
Apr. 2020 - Mar. 2024	BSc Kyushu Institute of Technology Department of Artificial Intelligence, Computer Science and Systems Engineering <u>Thesis: "Proposal of Representation-Conditional Latent Diffusion Models: Hematoma Classification in Brain CT Images"</u> GPA: 3.7 / 4.0
Apr. 2017 - Mar. 2020	High School Diploma Kanonji Daiichi High School Science and Mathematics Course GPA: 4.9 / 5.0

WORK EXPERIENCE

Dec. 2023 - Mar. 2026	Research Assistant Supervisor: Tsuyoshi Okita Researched and Developed foundation models
Apr. 2024 - Jun. 2026	Teaching Assistant Lecturer: Tsuyoshi Okita - Deep Learning(深層学習) - Deep Learning Exercise(知能情報工学プロジェクト-深層学習演習) - Deep Learning Advanced(深層学習特論I-大学院科目)
Oct. 2023 - Nov. 2023	OPTiM Corporation Research Intern / Part-time Research Assistant Engaged in research and developed AI technologies for agricultural DX

PUBLICATIONS

- [1]

Efficient Inference for Flow Matching via Unified Path CFG

Kosuke Ukita, Tsuyoshi Okita

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026 · Hong Kong, China

Oral

refereed

[2]

High-Performance Self-Supervised Learning by Joint Training of Flow Matching

Kosuke Ukita, Tsuyoshi Okita

The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)

May 2026 · Tangier, Morocco

Spotlight

refereed

[3]

Foundation Models to Tackle Activity Recognition in Unknown Domain: Sussex-Huawei Locomotion Challenge 2025 Task 2

Tsuyoshi Okita, **Kosuke Ukita**, Asahi Miyazaki, Daichi Kubota, Jukichi Ota, Naoki Kagiyaama, Asahi Nishikawa, Nagayasu Daichi, Syunya Tomitaka, Daisuke Nozaki, Yuki Odo, Raku Yamashita, Xiaolong Ye, Huayu Gao, Kazuki Okahashi, Koki Matsuishi, Masaharu Kagiyaama, Kodai Hirata, Haruki Kai, Lin Wang, Hristijan Gjoreski, Mathias Ciliberto, Paula Lago, Kazuya Murao, Daniel Roggen

Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25), pp. 983–991

October 2025 · Espoo, Finland

Oral

refereed

[4] **Multimodal Foundation Model for Cross-Modal Retrieval and Activity Recognition Tasks**

Koki Matsuishi, **Kosuke Ukita**, Tsuyoshi Okita

International Journal of Activity and Behavior Computing (IJABC), Care XDX Center, Kyushu Institute of Technology, Vol. 2025, No. 3, pp. 1-25

November 2025 · Kitakyushu, Japan

Oral

refereed

[5] **Energy-Efficient High-Performance Represen-tation Learning for Sensor Data via Flow Matching**

Kosuke Ukita, Tsuyoshi Okita

The 7th International Symposium on Neuromorphic AI Hardware, Paper No. P-19

October 2025 · Kitakyushu, Japan

Poster

non-refereed

[6] **Image Classification Using a Diffusion Model as a PreTraining Model**

Kosuke Ukita, Xiaolong Ye, Tsuyoshi Okita

arXiv preprint arXiv:2505.06890

May 2025

Preprint

non-refereed

[7] **AURA-MFM: 詳細な行動理解のためのCLIPを用いたマルチモーダル基盤モデルの構築**

松石康希, **浮田嵩祐**, 大北剛

情報処理学会研究報告(IPSJ SIG Technical Report), Vol. 2025-UBI-85, No. 8

February 2025 · Osaka, Japan

Oral

non-refereed

[8] **Improving Human Activity Recognition through Diffusion Models**

Kosuke Ukita, Tsuyoshi Okita

The International Symposium on Applied Engineering and Sciences (SAES2024), No. p0052

November 2024 · Kitakyushu, Japan

Oral

non-refereed

[9] **Ensemble Learning of Models for Human Activity Recognition**

Asahi Miyazaki, **Kosuke Ukita**, Tsuyoshi Okita

The International Symposium on Applied Engineering and Sciences (SAES2024), No. p0053

November 2024 · Kitakyushu, Japan

Oral

non-refereed

[10] **Analysis of Human Activity Recognition by Diffusion Models**

Kosuke Ukita, Tsuyoshi Okita

Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24), pp. 458–463

October 2024 · Melbourne, Australia

Oral

refereed

[11]

表現条件付き潜在拡散モデルと表現学習

浮田嵩祐, Ye Xiaolong, 大北剛

情報処理学会研究報告, 火の国情報シンポジウム 2024, No. A4-4

March 2024 · Kagoshima, Japan

Oral

non-refereed

[12]

Towards LLMs for Sensor Data: Multi-Task Self-Supervised Learning

Tsuyoshi Okita, Kosuke Ukita, Koki Matsuishi, Masaharu Kagiya, Kodai Hirata, Asahi Miyazaki

Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), pp. 499–504

October 2023 · Cancun, Mexico

Oral

refereed

GRANTS

- Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

Jun. 2026 · USD \$500
- International Travel Grant (海外渡航旅費援助)

The Telecommunications Advancement Foundation (電気通信普及財団)

Apr. 2026 · JPY ¥290,000
- Kyushu Institute of Technology SPRING Scholarship Awarde

Japan Science and Technology Agency (JST)

Apr. 2026 - Present · JPY ¥1,800,000 (+¥900,000) / year

AWARDS

- Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

Jun. 2026
- Spotlight Presentation (Top 3%)

The 29th Annual Conference on Artificial Intelligence and Statistics (AISTATS 2026)

May. 2026
- Incentive Award (奨励賞)

IPSJ Kyushu Branch (情報処理学会九州支部)

Mar. 2024
- Highest Award (最優秀賞)

The Japan Statistical Society Special Committee of Statistical Education, The 8th Sports Data Analysis Competition

May. 2019

SKILLS

LANGUAGES

Python · C++ · C · Java · JavaScript · TypeScript · Go · SQL · R

FRAMEWORKS & TOOLS

PyTorch · TensorFlow · Hugging Face · Jupyter · ABCI · Docker · Git

OTHER SKILLS

Deep Learning · Neural Networks · Model Training · Data Processing · API Development · Linux